

Giuseppe Iannaccone

professor

基本信息

姓名: Giuseppe Iannaccone

所属院校: 比萨大学

地区: Italy

年龄范围: 50岁以上

荣誉称号

★ IEEE Fellow

★ APS Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

电子科学与技术

材料物理

集成电路设计与集成系统

研究领域

纳米电子与先进工艺

微纳电子与纳米技术

半导体器件物理与工艺

低维材料与量子效应

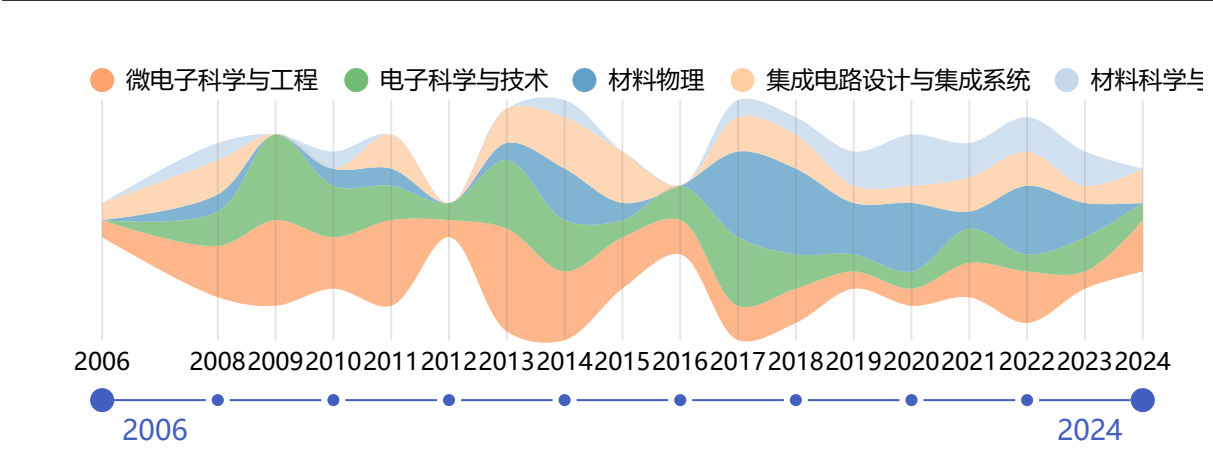
微电子器件与制造

个人简介

Giuseppe Iannaccone is a professor of Electronics at the University of Pisa, where he graduated in Electrical Engineering in 1992 and obtained a PhD in Information Engineering in 1996. He is the Chairman of IU.NET and has teaching activities in electronics and related fields. His research focuses on nanoelectronics and has expanded to areas like ultra - low - power circuits and AI. He coordinated multiple European and National projects, is a fellow of IEEE and APS, and an auditor of international funding agencies. With over 250

publications and high citations, he promotes science dissemination, is involved in technology transfer, founded a spin - off, and chaired a patent promotion commission. [https://www.iannaccone.org/]

近年发表的论文



Conditions for enhanced shot noise in field-effect transistors

PHYSICAL REVIEW APPLIED

Mazziotti, Fabrizio; Logoteta, Demetrio; Iannaccone, Giuseppe

被引用次数: 0 2024

CVD graphene contacts for lateral heterostructure MoS2 field effect transistors

NPJ 2D MATERIALS AND APPLICATIONS

Schneider, Daniel S.; Lucchesi, Leonardo; Reato, Eros; Wang, Zhenyu; Piacentini, Agata; Bolten, Jens; Marian, Damiano; Marin, Enrique G.; Radenovic, Aleksandra; Wang, Zhenxing; Fiori, Gianluca; Kis, Andras; Iannaccone, Giuseppe; Neumaier, Daniel; Lemme, Max C.

被引用次数: 2 2024

High-Precision Materials Printer for Fast Prototyping of Electronic Devices Based on 2D Materials

ADVANCED MATERIALS TECHNOLOGIES

Sargeni, Riccardo; Dimaggio, Elisabetta; Pieri, Francesco; Fabbri, Filippo; Pennelli, Giovanni; Colombo, Luigi; Iannaccone, Giuseppe; Macucci, Massimo; Fiori, Gianluca

被引用次数: 1 2024

Design Criteria of High-Temperature Integrated Circuits Using Standard SOI CMOS Process up to 300degree celsius

IEEE ACCESS

Sbrana, Christian; Catania, Alessandro; Paliy, Maksym; Di Pascoli, Stefano; Strangio, Sebastiano; Macucci, Massimo; Iannaccone, Giuseppe

被引用次数: 0 2024

Data Loggers for High-Temperature Industrial Environments

IEEE ACCESS

Cecchi, Roberto; Catania, Alessandro; Sbrana, Christian; Macucci, Massimo; Strangio, Sebastiano; Iannaccone, Giuseppe

被引用次数: 0

2024

Rhombohedral-stacked bilayer transition metal dichalcogenides for high-performance atomically thin CMOS devices

SCIENCE ADVANCES

Li, Xuefei; Shi, Xinhang; Marian, Damiano; Soriano, David; Cusati, Teresa; Iannaccone, Giuseppe; Fiori, Gianluca; Guo, Qi; Zhao, Wenjie; Wu, Yanqing

被引用次数: 15

2023

Printed transistors made of 2D material-based inks

NATURE REVIEWS MATERIALS

Conti, Silvia; Calabrese, Gabriele; Parvez, Khaled; Pimpolari, Lorenzo; Pieri, Francesco; Iannaccone, Giuseppe; Casiraghi, Cinzia; Fiori, Gianluca

被引用次数: 38

2023

Substitutional p-Type Doping in NbS₂-MoS₂ Lateral Heterostructures Grown by MOCVD

ADVANCED MATERIALS

Wang, Zhenyu; Tripathi, Mukesh; Golsanamlou, Zahra; Kumari, Poonam; Lovarelli, Giuseppe; Mazziotti, Fabrizio; Logoteta, Demetrio; Fiori, Gianluca; Sementa, Luca; Marega, Guilherme Migliato; Ji, Hyun Goo; Zhao, Yanfei; Radenovic, Aleksandra; Iannaccone, Giuseppe; Fortunelli, Alessandro; Kis, Andras

被引用次数: 27

2023

A 2-D-Material FET Verilog-A Model for Analog Neuromorphic Circuit Design

IEEE TRANSACTIONS ON ELECTRON DEVICES

Dubey, Prabhat Kumar; Strangio, Sebastiano; Marin, Enrique G.; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 0

2023

Load Modulation Feedback in Adaptive Matching Networks for Low-Coupling Wireless Power Transfer Systems

ELECTRONICS

Bertozzi, Michele; Catania, Alessandro; Bandini, Gabriele; Strangio, Sebastiano; Iannaccone, Giuseppe

被引用次数: 1

2023

Multi-scale simulations of two dimensional material based devices: the NanoTCAD ViDES suite

JOURNAL OF COMPUTATIONAL ELECTRONICS

Marian, Damiano; Marin, Enrique G.; Perucchini, Marta; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 7

2023

Multiscale Pseudoatomistic Quantum Transport Modeling for van der Waals Heterostructures

PHYSICAL REVIEW APPLIED

Lovarelli, Giuseppe; Calogero, Gaetano; Fiori, Gianluca; Iannaccone, Giuseppe

被引用次数: 7

2022

(La,Ba)SnO₃-based Thin-Film Transistors: Large-Signal Model and Scaling Projections

PHYSICAL REVIEW APPLIED

Mazziotti, Fabrizio; Logoteta, Demetrio; Iannaccone, Giuseppe

被引用次数: 1

2022

Assessment of paper-based MoS₂ FET for Physically Unclonable Functions

SOLID-STATE ELECTRONICS

Vatalaro, Massimo; De Rose, Raffaele; Lanuzza, Marco; Magnone, Paolo; Conti, Silvia; Iannaccone, Giuseppe; Crupi, Felice

被引用次数: 4

2022

Low-Power Artificial Neural Network Perceptron Based on Monolayer MoS₂

ACS NANO

Marega, Guilherme Migliato; Wang, Zhenyu; Paliy, Maksym; Giusi, Gino; Strangio, Sebastiano; Castiglione, Francesco; Callegari, Christian; Tripathi, Mukesh; Radenovic, Aleksandra; Iannaccone, Giuseppe; Kis, Andras

被引用次数: 24

2022

Vertical Heterostructures between Transition-Metal Dichalcogenides-A Theoretical Analysis of the NbS₂/WSe₂ Junction

ADVANCED ELECTRONIC MATERIALS

Golsanamlou, Zahra; Kumari, Poonam; Sementa, Luca; Cusati, Teresa; Iannaccone, Giuseppe; Fortunelli, Alessandro

被引用次数: 6

2022

Electronic Transport in 2D-Based Printed FETs from a Multiscale Perspective

ADVANCED ELECTRONIC MATERIALS

Perucchini, Marta; Marian, Damiano; Marin, Enrique G.; Cusati, Teresa; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 9

2022

All-Analog Silicon Integration of Image Sensor and Neural Computing Engine for Image Classification

IEEE ACCESS

Zambrano, Benjamin; Strangio, Sebastiano; Rizzo, Tommaso; Garzon, Esteban; Lanuzza, Marco; Iannaccone, Giuseppe

被引用次数: 4

2022

Time Domain Analog Neuromorphic Engine Based on High-Density Non-Volatile Memory in Single-Poly CMOS

IEEE ACCESS

Rizzo, Tommaso; Strangio, Sebastiano; Iannaccone, Giuseppe

被引用次数: 7

2022

A 0.6V-1.8V Compact Temperature Sensor With 0.24 °C Resolution, ±1.4 °C Inaccuracy and 1.06nJ per Conversion

IEEE SENSORS JOURNAL

Zambrano, Benjamin; Garzon, Esteban; Strangio, Sebastiano; Iannaccone, Giuseppe; Lanuzza, Marco

被引用次数: 9

2022

A 6.78 MHz Maximum Efficiency Tracking Active Rectifier with Load Modulation Control for Wireless Power Transfer to Implantable Medical Devices

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Rizzo, Tommaso; Catania, Alessandro; Bertolacci, Gabriele; Strangio, Sebastiano; Iannaccone, Giuseppe

被引用次数: 1

2022

Assessment of 2D-FET Based Digital and Analog Circuits on Paper

SOLID-STATE ELECTRONICS

Vatalaro, Massimo; De Rose, Raffaele; Lanuzza, Marco; Iannaccone, Giuseppe; Crupi, Felice

被引用次数: 2

2021

Single-poly floating-gate memory cell options for analog neural networks

SOLID-STATE ELECTRONICS

Paliy, Maksym; Rizzo, Tommaso; Ruiiu, Piero; Strangio, Sebastiano; Iannaccone, Giuseppe

被引用次数: 0

2021

Inkjet-printed low-dimensional materials-based complementary electronic circuits on paper

NPJ 2D MATERIALS AND APPLICATIONS

Brunetti, Irene; Pimpolari, Lorenzo; Conti, Silvia; Worsley, Robyn; Majee, Subimal; Polyushkin, Dmitry K.; Paur, Matthias; Dimaggio, Elisabetta; Pennelli, Giovanni; Iannaccone, Giuseppe; Macucci, Massimo; Pieri, Francesco; Mueller, Thomas; Casiraghi, Cinzia; Fiori, Gianluca

被引用次数: 20

2021

1/f Noise Characterization of Bilayer MoS₂ Field-Effect Transistors on Paper with Inkjet-Printed Contacts and hBN Dielectrics

ADVANCED ELECTRONIC MATERIALS

Pimpolari, Lorenzo; Calabrese, Gabriele; Conti, Silvia; Worsley, Robyn; Majee, Subimal; Polyushkin, Dmitry K.; Paur, Matthias; Casiraghi, Cinzia; Mueller, Thomas; Iannaccone, Giuseppe; Macucci, Massimo; Fiori, Gianluca

被引用次数: 6

2021

Power Electronics Based on Wide-Bandgap Semiconductors: Opportunities and Challenges

IEEE ACCESS

Iannaccone, Giuseppe; Sbrana, Christian; Morelli, Iacopo; Strangio, Sebastiano

被引用次数: 49

2021

Physical insights on transistors based on lateral heterostructures of monolayer and multilayer PtSe₂ via Ab initio modelling of interfaces

SCIENTIFIC REPORTS

Calogero, Gaetano; Marian, Damiano; Marin, Enrique G.; Fiori, Gianluca; Iannaccone, Giuseppe

被引用次数: 4

2021

Assessment of Two-Dimensional Materials-Based Technology for Analog Neural Networks

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Paliy, Maksym; Strangio, Sebastiano; Ruiiu, Piero; Iannaccone, Giuseppe

被引用次数: 3

2021

Electric-field controlled spin transport in bilayer CrI3

IEEE 51ST EUROPEAN SOLID-STATE DEVICE RESEARCH CONFERENCE (ESSDERC 2021)

Marian, Damiano; Soriano, David; Marin, Enrique G.; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 0

2021

Electric-field controlled spin transport in bilayer CrI3

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Marian, Damiano; Soriano, David; Marin, Enrique G.; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 0

2021

Sub-Maxwellian Source Injection and Negative Differential Transconductance in Decorated Graphene Nanoribbons

PHYSICAL REVIEW APPLIED

Marian, Damiano; Marin, Enrique G.; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 2

2020

Lateral Heterostructure Field-Effect Transistors Based on Two-Dimensional Material Stacks with Varying Thickness and Energy Filtering Source

ACS NANO

Marin, Enrique G.; Marian, Damiano; Perucchini, Marta; Fiori, Gianluca; Iannaccone, Giuseppe

被引用次数: 50

2020

Inkjet-printed graphene Hall mobility measurements and low-frequency noise characterization

NANOSCALE

Calabrese, Gabriele; Pimpolari, Lorenzo; Conti, Silvia; Mavier, Fabrice; Majee, Subimal; Worsley, Robyn; Wang, Zihao; Pieri, Francesco; Basso, Giovanni; Pennelli, Giovanni; Parvez, Khaled; Brooks, David; Macucci, Massimo; Iannaccone, Giuseppe; Novoselov, Kostya S.; Casiraghi, Cinzia; Fiori, Gianluca

被引用次数: 19

2020

Analog Vector-Matrix Multiplier Based on Programmable Current Mirrors for Neural Network Integrated Circuits

IEEE ACCESS

Paliy, Maksym; Strangio, Sebastiano; Ruiiu, Piero; Rizzo, Tommaso; Iannaccone, Giuseppe

被引用次数: 17

2020

Analogue two-dimensional semiconductor electronics

NATURE ELECTRONICS

Polyushkin, Dmitry K.; Wachter, Stefan; Mennel, Lukas; Paur, Matthias; Paliy, Maksym; Iannaccone, Giuseppe; Fiori, Gianluca; Neumaier, Daniel; Canto, Barbara; Mueller, Thomas

被引用次数: 89

2020

Theoretical Analysis of a 2D Metallic/Semiconducting Transition-Metal Dichalcogenide NbS₂/WSe₂ Hybrid Interface

ADVANCED THEORY AND SIMULATIONS

Golsanamlou, Zahra; Sementa, Luca; Cusati, Teresa; Iannaccone, Giuseppe; Fortunelli, Alessandro

被引用次数: 5

2020

Low-voltage 2D materials-based printed field-effect transistors for integrated digital and analog electronics on paper

NATURE COMMUNICATIONS

Conti, Silvia; Pimpolari, Lorenzo; Calabrese, Gabriele; Worsley, Robyn; Majee, Subimal; Polyushkin, Dmitry K.; Paur, Matthias; Pace, Simona; Keum, Dong Hoon; Fabbri, Filippo; Iannaccone, Giuseppe; Macucci, Massimo; Coletti, Camilla; Mueller, Thomas; Casiraghi, Cinzia; Fiori, Gianluca

被引用次数: 132

2020

All-2D Material Inkjet-Printed Capacitors: Toward Fully Printed Integrated Circuits

ACS NANO

Worsley, Robyn; Pimpolari, Lorenzo; McManus, Daryl; Ge, Ning; Ionescu, Robert; Wittkopf, Jarrid A.; Alieva, Adriana; Basso, Giovanni; Macucci, Massimo; Iannaccone, Giuseppe; Novoselov, Kostya S.; Holder, Helen; Fiori, Gianluca; Casiraghi, Cinzia

被引用次数: 101

2019

Phonon-assisted carrier transport through a lattice-mismatched interface

NPG ASIA MATERIALS

Yoon, Hyong Seo; Oh, Juyeong; Park, Jae Young; Kang, JeongSeob; Kwon, Junyoung; Cusati, Teresa; Fiori, Gianluca; Iannaccone, Giuseppe; Fortunelli, Alessandro; Ozelik, V. Ongun; Lee, Gwan-Hyoung; Low, Tony; Jun, Seong Chan

被引用次数: 11

2019

Flexible One-Dimensional Metal-Insulator-Graphene Diode

ACS APPLIED ELECTRONIC MATERIALS

Wang, Zhenxing; Uzlu, Burak; Shaygan, Mehrdad; Otto, Martin; Ribeiro, Mario; Marin, Enrique Gonzalez; Iannaccone, Giuseppe; Fiori, Gianluca; Elsayed, Mohamed Saeed; Negra, Renato; Neumaier, Daniel

被引用次数: 24

2019

Ultralow Specific Contact Resistivity in Metal-Graphene Junctions via Contact Engineering

ADVANCED MATERIALS INTERFACES

Passi, Vikram; Gahoi, Amit; Marin, Enrique G.; Cusati, Teresa; Fortunelli, Alessandro; Iannaccone, Giuseppe; Fiori, Gianluca; Lemme, Max C.

被引用次数: 44

2019

A critical review of reduced one-dimensional beam models of piezoelectric composite beams

JOURNAL OF INTELLIGENT MATERIAL SYSTEMS AND STRUCTURES

Luschi, Luca; Iannaccone, Giuseppe; Pieri, Francesco

被引用次数: 7

2019

A Steep-Slope MoS₂-Nanoribbon MOSFET Based on an Intrinsic Cold-Contact Effect

IEEE ELECTRON DEVICE LETTERS

Logoteta, Demetrio; Pala, Marco G.; Choukroun, Jean; Dollfus, Philippe; Iannaccone, Giuseppe

被引用次数: 28

2019

Physical insights into the operation of a 1-nm gate length transistor based on MoS2 with metallic carbon nanotube gate

APPLIED PHYSICS LETTERS

Perucchini, Marta; Marin, Enrique G.; Marian, Damiano; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 5

2018

Stacking and interlayer electron transport in MoS2

PHYSICAL REVIEW B

Cusati, Teresa; Fortunelli, Alessandro; Fiori, Gianluca; Iannaccone, Giuseppe

被引用次数: 11

2018

Quantum engineering of transistors based on 2D materials heterostructures

NATURE NANOTECHNOLOGY

Iannaccone, Giuseppe; Bonaccorso, Francesco; Colombo, Luigi; Fiori, Gianluca

被引用次数: 363

2018

Reconfigurable Diodes Based on Vertical WSe2 Transistors with van der Waals Bonded Contacts

ADVANCED MATERIALS

Avsar, Ahmet; Marinov, Kolyo; Marin, Enrique Gonzalez; Iannaccone, Giuseppe; Watanabe, Kenji; Taniguchi, Takashi; Fiori, Gianluca; Kis, Andras

被引用次数: 37

2018

First-Principles Simulations of FETs Based on Two-Dimensional InSe

IEEE ELECTRON DEVICE LETTERS

Marin, Enrique G.; Marian, Damiano; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 39

2018

Threshold Voltage Instability in p-GaN Gate AlGaN/GaN HFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sayadi, Luca; Iannaccone, Giuseppe; Sicre, Sebastien; Haeberlen, Oliver; Curatola, Gilberto

被引用次数: 174

2018

The Role of Silicon Substrate on the Leakage Current Through GaN-on-Si Epitaxial Layers

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sayadi, Luca; Iannaccone, Giuseppe; Haeberlen, Oliver; Fiori, Gianluca; Tomberger, Manuel; Knuuttila, Lauri O.; Curatola, Gilberto

被引用次数: 24

2018

A portable class of 3-transistor current references with low-power sub-0.5V operation

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Crupi, Felice; De Rose, Raffaele; Paliy, Maksym; Lanuzza, Marco; Perna, Mattia; Iannaccone, Giuseppe

Transistor Concepts Based on Lateral Heterostructures of Metallic and Semiconducting Phases of MoS₂

PHYSICAL REVIEW APPLIED

Marian, Damiano; Dib, Elias; Cusati, Teresa; Marin, Enrique G.; Fortunelli, Alessandro; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 41

2017

Water-based and biocompatible 2D crystal inks for all-inkjet-printed heterostructures

NATURE NANOTECHNOLOGY

McManus, Daryl; Vranic, Sandra; Withers, Freddie; Sanchez-Romaguera, Veronica; Macucci, Massimo; Yang, Huafeng; Sorrentino, Roberto; Parvez, Khaled; Son, Seok-Kyun; Iannaccone, Giuseppe; Kostarelos, Kostas; Fiori, Gianluca; Casiraghi, Cinzia

被引用次数: 480

2017

High-Performance 2D p-Type Transistors Based on GaSe Layers: An Ab Initio Study

ADVANCED ELECTRONIC MATERIALS

Kuc, Agnieszka; Cusati, Teresa; Dib, Elias; Oliveira, Augusto F.; Fortunelli, Alessandro; Iannaccone, Giuseppe; Heine, Thomas; Fiori, Gianluca

被引用次数: 25

2017

High performance metal-insulator-graphene diodes for radio frequency power detection application

NANOSCALE

Shaygan, Mehrdad; Wang, Zhenxing; Elsayed, Mohamed Saeed; Otto, Martin; Iannaccone, Giuseppe; Ghareeb, Ahmed Hamed; Fiori, Gianluca; Negra, Renato; Neumaier, Daniel

被引用次数: 36

2017

Growth-Induced Strain in Chemical Vapor Deposited Monolayer MoS₂: Experimental and Theoretical Investigation

ADVANCED MATERIALS INTERFACES

Kataria, Satender; Wagner, Stefan; Cusati, Teresa; Fortunelli, Alessandro; Iannaccone, Giuseppe; Pandey, Himadri; Fiori, Gianluca; Lemme, Max C.

被引用次数: 58

2017

GaN Nanowire n-MOSFET With 5 nm Channel Length for Applications in Digital Electronics

IEEE ELECTRON DEVICE LETTERS

Chowdhury, Nadim; Iannaccone, Giuseppe; Fiori, Gianluca; Antoniadis, Dimitri A.; Palacios, Tomas

被引用次数: 40

2017

An Ultralow-Voltage Energy-Efficient Level Shifter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lanuzza, Marco; Crupi, Felice; Rao, Sandro; De Rose, Raffaele; Strangio, Sebastiano; Iannaccone, Giuseppe

被引用次数: 70

2017

Temperature Compensation of Silicon Lamé Resonators Using Etch Holes: Theory and Design Methodology

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Luschi, Luca; Iannaccone, Giuseppe; Pieri, Francesco

被引用次数: 7

2017

Low energy/delay overhead level shifter for wide-range voltage conversion

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Lanuzza, Marco; Crupi, Felice; Rao, Sandro; De Rose, Raffaele; Iannaccone, Giuseppe

被引用次数: 4

2017

On current transients in MoS₂ Field Effect Transistors

SCIENTIFIC REPORTS

Macucci, Massimo; Tambellini, Gerry; Ovchinnikov, Dmitry; Kis, Andras; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 4

2017

Electrical properties of graphenemetal contacts

SCIENTIFIC REPORTS

Cusati, Teresa; Fiori, Gianluca; Gahoi, Amit; Passi, Vikram; Lemme, Max C.; Fortunelli, Alessandro; Iannaccone, Giuseppe

被引用次数: 133

2017

Material-Device-Circuit Cooptimization of 2D Material based FETs for Ultra-Scaled Technology Nodes

SCIENTIFIC REPORTS

Agarwal, Tarun Kumar; Soree, Bart; Radu, Iuliana; Raghavan, Praveen; Iannaccone, Giuseppe; Fiori, Gianluca; Dehaene, Wim; Heyns, Marc

被引用次数: 17

2017

Comparison of short-channel effects in monolayer MoS₂ based junctionless and inversion-mode field-effect transistors

APPLIED PHYSICS LETTERS

Agarwal, Tarun; Soree, Bart; Radu, Iuliana; Raghavan, Praveen; Fiori, Gianluca; Iannaccone, Giuseppe; Thean, Aaron; Heyns, Marc; Dehaene, Wim

被引用次数: 22

2016

Low-Power Wearable ECG Monitoring System for Multiple-Patient Remote Monitoring

IEEE SENSORS JOURNAL

Spano, Elisa; Di Pascoli, Stefano; Iannaccone, Giuseppe

被引用次数: 138

2016

Performance of arsenene and antimonene double-gate MOSFETs from first principles

NATURE COMMUNICATIONS

Pizzi, Giovanni; Gibertini, Marco; Dib, Elias; Marzari, Nicola; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 280

2016

Can graphene outperform indium tin oxide as transparent electrode in organic solar cells?

2D MATERIALS

Paletti, Paolo; Pawar, Ravinder; Ulisse, Giacomo; Brunetti, Francesca; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 9

2015

Last-Meter Smart Grid Embedded in an Internet-of-Things Platform

IEEE TRANSACTIONS ON SMART GRID

Spano, Elisa; Niccolini, Luca; Di Pascoli, Stefano; Iannaccone, Giuseppe

被引用次数: 119

2015

A Sub-kT/q Voltage Reference Operating at 150 mV

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Albano, Domenico; Crupi, Felice; Cucchi, Francesca; Iannaccone, Giuseppe

被引用次数: 62

2015

A sub-1V nanopower temperature-compensated sub-threshold CMOS voltage reference with 0.065%/V line sensitivity

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Magnelli, Luca; Crupi, Felice; Corsonello, Pasquale; Iannaccone, Giuseppe

被引用次数: 5

2015

A SPICE-Compatible Model of MOS-Type Graphene Nano-Ribbon Field-Effect Transistors Enabling Gate-and Circuit-Level Delay and Power Analysis Under Process Variation

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Chen, Ying-Yu; Sangai, Amit; Rogachev, Artem; Gholipour, Morteza; Iannaccone, Giuseppe; Fiori, Gianluca; Chen, Deming

被引用次数: 49

2015

Suppressed and enhanced shot noise in one dimensional field-effect transistors

JOURNAL OF COMPUTATIONAL ELECTRONICS

Iannaccone, Giuseppe; Betti, Alessandro; Fiori, Gianluca

被引用次数: 7

2015

Electronics based on two-dimensional materials

NATURE NANOTECHNOLOGY

Fiori, Gianluca; Bonaccorso, Francesco; Iannaccone, Giuseppe; Palacios, Tomas; Neumaier, Daniel; Seabaugh, Alan; Banerjee, Sanjay K.; Colombo, Luigi

被引用次数: 2770

2014

On Transport in Vertical Graphene Heterostructures

IEEE ELECTRON DEVICE LETTERS

Zhang, Qin; Fiori, Gianluca; Iannaccone, Giuseppe

被引用次数: 15

2014

Two-Dimensional Tunnel Transistors Based on Bi₂Se₃ Thin Film

IEEE ELECTRON DEVICE LETTERS

Zhang, Qin; Iannaccone, Giuseppe; Fiori, Gianluca

被引用次数: 30

2014

Simulation of the Performance of Graphene FETs With a Semiclassical Model, Including Band-to-Band Tunneling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Paussa, Alan; Fiori, Gianluca; Palestri, Pierpaolo; Geromel, Matteo; Esseni, David; Iannaccone, Giuseppe; Selmi, Luca

被引用次数: 17

2014

Bilayer Graphene Transistors for Analog Electronics

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fiori, Gianluca; Neumaier, Daniel; Szafranek, Bart N.; Iannaccone, Giuseppe

被引用次数: 39

2014

An Open-Source Multiscale Framework for the Simulation of Nanoscale Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Bruzzone, Samantha; Iannaccone, Giuseppe; Marzari, Nicola; Fiori, Gianluca

被引用次数: 54

2014

A picopower temperature-compensated, subthreshold CMOS voltage reference

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Albano, Domenico; Crupi, Felice; Cucchi, Francesca; Iannaccone, Giuseppe

被引用次数: 24

2014

Design of a 75-nW, 0.5-V subthreshold complementary metal-oxide-semiconductor operational amplifier

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Magnelli, Luca; Amoroso, Francesco A.; Crupi, Felice; Cappuccino, Gregorio; Iannaccone, Giuseppe

被引用次数: 52

2014

Heterojunction Hybrid Devices from Vapor Phase Grown MoS₂

SCIENTIFIC REPORTS

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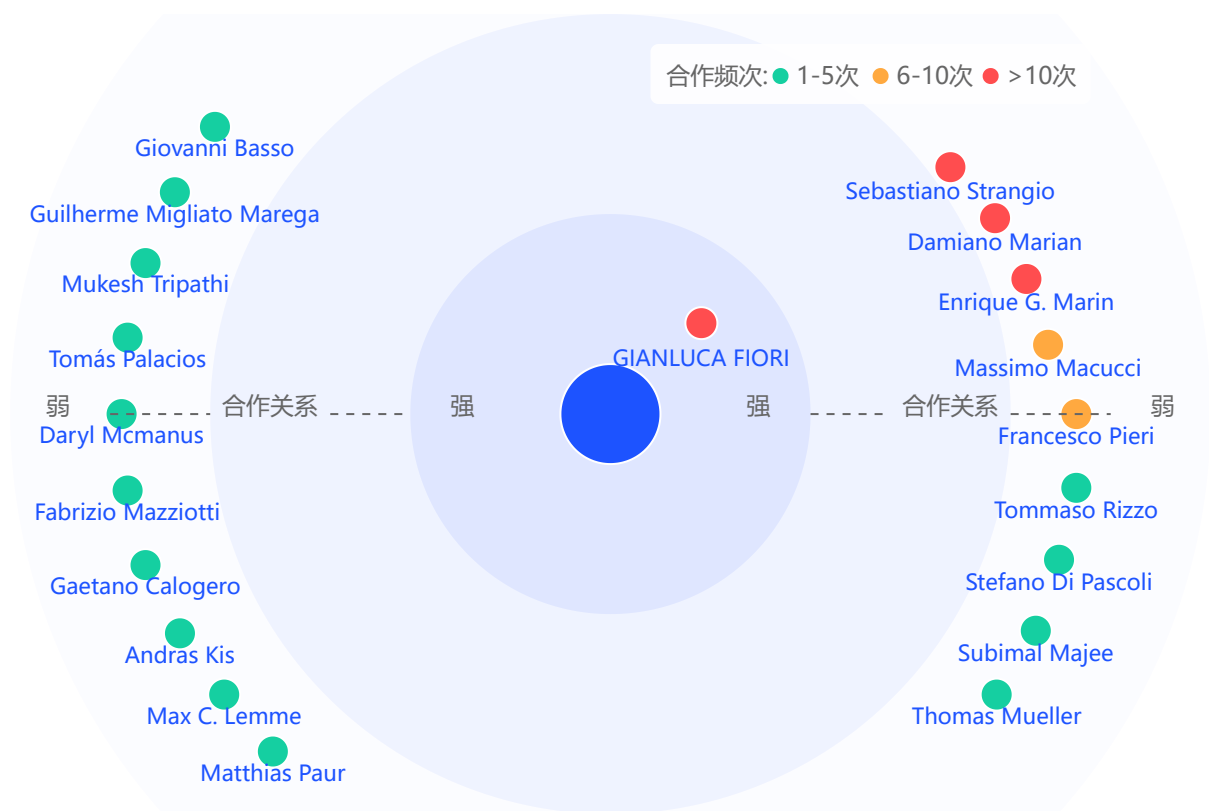
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